

AMIT KUMAR

M.Sc. Electronics | Semiconductor Device Physics | IC Design Fundamentals | VLSI (Learning)
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OBJECTIVE

M.Sc. Electronics graduate with a strong academic foundation in semiconductor device physics, CMOS fundamentals, and digital/analog IC design principles, seeking to begin an IC Design engineering career at Silicon Labs. Hands-on research experience in thin-film fabrication and precision semiconductor material characterization has built rigorous, process-aware technical thinking directly transferable to IC design and verification workflows. Eager to grow under experienced mentorship in a low-power wireless SoC design environment.

EDUCATION

M.Sc. Electronics, Department of Electronic Science, University of Delhi South Campus 2023 – 2025

CGPA: 7.27 / 10 · Relevant Coursework: Semiconductor Devices, VLSI Technology, Digital Circuit Design, Analog Circuit Design, Fabrication & Characterization Techniques

B.Sc. (Hons.) Electronic Science, Acharya Narendra Dev College, University of Delhi 2020 – 2023

CGPA: 7.43 / 10 · Key Subjects: Digital Electronics & VHDL, Semiconductor Devices, Microprocessors (8085/8051), Signals & Systems

RELEVANT SKILLS

- **IC Design Fundamentals:** Semiconductor Device Physics (MOSFETs, BJTs, diodes), CMOS circuit fundamentals, digital & analog circuit design, basic timing concepts — coursework-level foundation, actively deepening through self-study and VLSI certification
- **Digital Design:** Digital Circuit Design, Verilog (beginner), basic VLSI concepts — NIELIT VLSI for Beginners certified
- **Semiconductor Materials & Process:** Hands-on experience with thin-film deposition (DC sputtering) and material characterization (XRD, VSM, FMR) — direct exposure to semiconductor process discipline and precision measurement
- **Simulation & Analysis Tools:** Proteus, Multisim, PSpice, MATLAB, Origin Pro — no direct exposure yet to industry IC design EDA tools (Cadence/Synopsys); comfortable and fast at learning new toolchains
- **Programming:** C, C++, Python (basic)

RESEARCH EXPERIENCE — M.SC. DISSERTATION

Thin Film Fabrication & Semiconductor Characterization Jan – May 2025

Department of Electronic Science, UDSC · Supervisor: Prof. Manoj Kumar Khanna

- Fabricated Fe₈₀Co₂₀ thin films (6–15 nm) via DC sputtering — controlling vacuum (~10⁻⁶ Torr), Ar pressure (3–5 mTorr), and deposition power (50–200 W) with process-level discipline directly relevant to semiconductor fabrication rigor.
- Applied XRD, VSM, and FMR characterization techniques to extract structural and magnetic parameters, independently learning and operating precision instrumentation with zero prior hands-on experience — demonstrating fast, reliable technical ramp-up ability.
- Interpreted multi-technique measurement data to draw consistent physical conclusions, reflecting the same analytical rigor required in IC design verification and Debug.

WORK EXPERIENCE

Electronics Intern, Sagedel Tech LLP Jul – Dec 2023

- Designed and tested analog/digital circuit modules (BJT, 555 Timer IC, Arduino); performed structured functional verification on prototype electronics kits — early exposure to circuit-level design and test discipline.

CERTIFICATIONS

- VLSI for Beginners — NIELIT Calicut, April 2025 (5-day online certification)
- UGC-NET, Electronic Science (Dec 2025) — Qualified for Ph.D. admission